

Skills Summary

Graphing Quadratics from Vertex or Standard Form

Use this reference while completing your worksheet or when studying.

Vertex form $y = a(x - h)^2 + k$ vertex (h, k) , axis $x = h$.

Standard form $y = ax^2 + bx + c$ axis $x = -\frac{b}{2a}$, y -intercept c ; substitute the axis value to get the vertex height.

What a does: $a > 0$ opens up (vertex is a minimum), $a < 0$ opens down (maximum); $|a| > 1$ narrower than $y = x^2$, $|a| < 1$ wider.

To sketch: plot the vertex, draw the axis, plot the y -intercept, mirror it across the axis, add the x -intercepts if there are any, and curve through them.

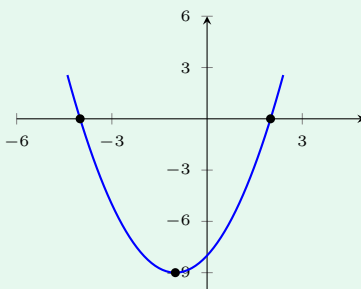
1. Graph from vertex form

Read h and k straight off the equation — but $(x + 1)^2$ means $(x - (-1))^2$, so $h = -1$. The sign inside the parentheses always flips. Then a tells you the direction and the width, and two symmetric points finish the curve.

Example: Sketch $y = (x + 1)^2 - 9$ and state the vertex.

Step 1. The equation is already in vertex form, so no algebra is needed — read h and k , then use symmetry for the rest of the curve.

Step 2. $(x + 1)^2 = (x - (-1))^2$ gives $h = -1$, and $k = -9$. Check: $y(-1) = 0 - 9 = -9$. Since $a = 1 > 0$ it opens up, and setting $y = 0$ gives $(x + 1)^2 = 9$, so the x -intercepts are -4 and 2 .



Vertex height at $x = -1$: -9

Try it: For $y = -2(x - 4)^2 + 3$, what is the y -value at the vertex, and does the parabola open up or down?

check: $a = -2$; opens down

2. Graph from standard form

The axis of symmetry comes first: $x = -\frac{b}{2a}$. Keep both minus signs — the one in the formula and any minus on b or on a . Substituting that x back into the equation gives the vertex height, and c is the y -intercept for free.

Example: Find the vertex of $y = x^2 - 4x + 1$.

Step 1. Nothing factors nicely here, so go through the axis of symmetry rather than hunting for intercepts.

Step 2. $a = 1$, $b = -4$, so $x = -\frac{-4}{2(1)} = 2$. Substituting: $y(2) = 4 - 8 + 1 = -3$. Vertex height at $x = 2$: -3 , so the vertex is $(2, -3)$ and the y -intercept is 1.

Try it: Find the vertex of $y = x^2 + 6x + 5$. (Axis first, then substitute.)

check: axis $x = -3$, vertex height -4

3. Use the x -intercepts

The x -intercepts are the solutions of $y = 0$. Factor when you can: from $(x - p)(x - q)$ the intercepts are p and q , and the axis of symmetry sits exactly midway between them at $x = \frac{p+q}{2}$ — an independent way to find the vertex, and a free check on $-\frac{b}{2a}$.

Example: Find the x -intercepts of $y = x^2 - 2x - 8$.

Step 1. The question asks where the graph meets the x -axis, which is the same as solving $y = 0$ — so factor rather than complete the square.

Step 2. Two numbers with product -8 and sum -2 are -4 and 2 : $x^2 - 2x - 8 = (x - 4)(x + 2)$, so $x - 4 = 0$ or $x + 2 = 0$. x -intercepts: $x = 4$ and $x = -2$ (so the axis is $x = 1$)

Try it: Find the x -intercepts of $y = x^2 - x - 6$, then say where the axis of symmetry is.

check: axis $x = \frac{1}{2}$, $x = 3$ and $x = -2$

(!) Watch out: $y = (x + 5)^2$ has its vertex at $x = -5$, not $x = 5$. And in $-\frac{b}{2a}$ the sign of a counts too: for $y = -x^2 + 4x + 5$ the axis is $x = -\frac{4}{-2} = 2$, not -2 .